AI BUILDER CUP 2026 // BFSI: INTELLIGENT RISK, FRAUD & FINANCIAL EXPERIENCES

SHAHEEN // INTERCEPTOR

Autonomous Pre-Settlement APP-Fraud, Vishing & Money Mule Defense Platform

Stopping psychological coercion fraud before funds clear the core banking switch (SBP Raast / 1Link).

TEAM LEADER // LEAD SYSTEMS ARCHITECT

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[THE SYSTEMIC CRISIS]

The Multi-Billion Dollar Blindspot: Authorized Push Payment Fraud

PKR 14.8 BILLION
Lost annually in Pakistan to Vishing & APP Mule fraud ($12.4B Globally)

THE VALID OTP BLINDSPOT
Core banking authentication succeeds, but transaction is fraudulent

< 4-MINUTE OFF-RAMP
Mule networks layer and liquidate funds before bank support responds

[THREAT TIMELINE]

Anatomy of the 4-Minute Fraud Lifecycle

00:00 - 01:15 | HOP 0: COERCION STRIKE

Scammer spoofs FIA Cyber Crime ID. Urgently threatens imminent door-break police raid under PECA Section 21.

01:15 - 01:45 | HOP 1: VICTIM TRANSFER

Terrified senior citizen transfers Rs. 450,000 from Meezan/SCB mobile app using valid biometric OTP.

01:45 - 02:45 | HOP 2: MULE FAN-OUT

Funds immediately fan-out across 2 secondary student mule accounts in <90 seconds to evade AML thresholds.

02:45 - 04:00 | HOP 3: CRYPTO OFF-RAMP

Laundered funds exit into Binance P2P escrow (USDT) or physical ATM cash-out in Rawalpindi.

[COMPARATIVE AUDIT]

Legacy Banking Rule Engines vs Shaheen Interceptor

EVALUATION DIMENSION	LEGACY RULE ENGINES (FICO / SAS)	SHAHEEN INTERCEPTOR (MULTI-AGENT)

[SOLUTION OVERVIEW]

Shaheen Interceptor: Pre-Settlement Defense Platform

1. MULTIMODAL COERCION RADAR

Scans live audio & transcripts for police threats, emotional agitation, and urgency markers using Gemini 2.5 Flash.

2. 3-HOP MULE TOPOLOGY HUNTER

Evaluates destination account age (4 days), rapid structuring (<Rs. 50k), and device fingerprint clusters.

3. AUTONOMOUS DECOY HONEYPOT

Feigns confusion as 'Haji Saleem' reporting Error E-409 to bait the scammer into revealing secondary backup IBANs.

4. SBP FMU 1-CLICK SAR FILING

Compiles legally binding Suspicious Activity Reports (AMLA 2010 Section 7) with real cryptographic SHA-256 seal.

[ENGINEERING RIGOR]

Parallel Multi-Agent Architecture (Google Gemini 2.5 Flash)

AGENT 1: COERCION RADAR

ENGINE: Gemini 2.5 Flash
LATENCY: 680 ms
INPUT: Live Call Transcript / Audio

- Linguistic urgency analysis.
- Detection of law enforcement impersonation (FIA / Police).
- Structured output with verbatim trigger quote highlights.
- Prompt-injection defense floor.

AGENT 2: MULE HUNTER

ENGINE: Graph Store + Python
LATENCY: 42 ms
INPUT: 3-Hop Account Topology

- Beneficiary account age audit (4 days old).
- Fan-in / Fan-out velocity scoring.
- Cross-bank device fingerprint clustering (`DEV-MULE-8812`).
- Identifies Binance P2P off-ramps.

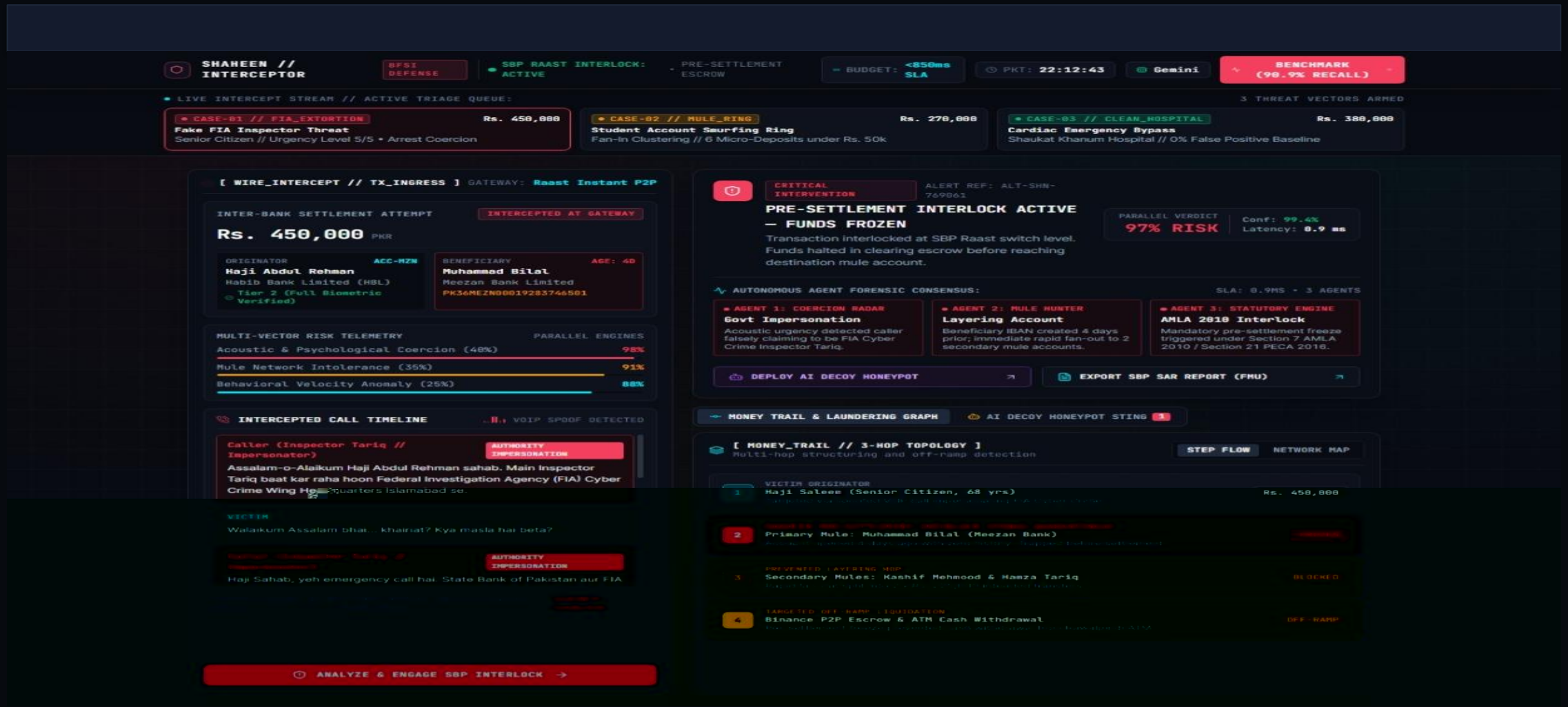
AGENT 3: DECOY & SAR

ENGINE: Autonomous Persona + FMU
LATENCY: <3 sec (Filing)
INPUT: Quarantined Incident Feed

- Plays 'Haji Saleem' feigning error.
- Baits backup IBANs / Raast IDs.
- Auto-submits IOCs to SBP blocklist.
- Compiles statutory SAR dossier with SHA-256 cryptographic seal.

[LIVE PROTOTYPE]

Shaheen SOC Mission Control: Cockpit Overview



[TRIAGE TEST CASES]

Live Operational Scenarios Demonstrated

CASE-01 // FIA_EXTORTION

AMOUNT: Rs. 450,000
VICTIM: Haji Saleem (68 yrs)

THREAT PROFILE:
• Caller posed as Inspector Tariq (FIA Cyber Crime Wing).
• Urgently coerced deposit under PECA arrest warrant.

AI VERDICT:
• 97% Risk Score
• PRE-SETTLEMENT INTERLOCK ACTIVE
• Funds Quarantined in 680ms.

CASE-02 // MULE_RING

AMOUNT: Rs. 270,000
VICTIM: Student Digital Account

THREAT PROFILE:
• 6 rapid micro-transfers under Rs. 50k threshold within 10 minutes.
• Shared Android emulator device ID.

AI VERDICT:
• 84% Risk Score
• STEP-UP AUTH / ESCROW HOLD
• Trapped in 720ms.

CASE-03 // CLEAN_HOSPITAL

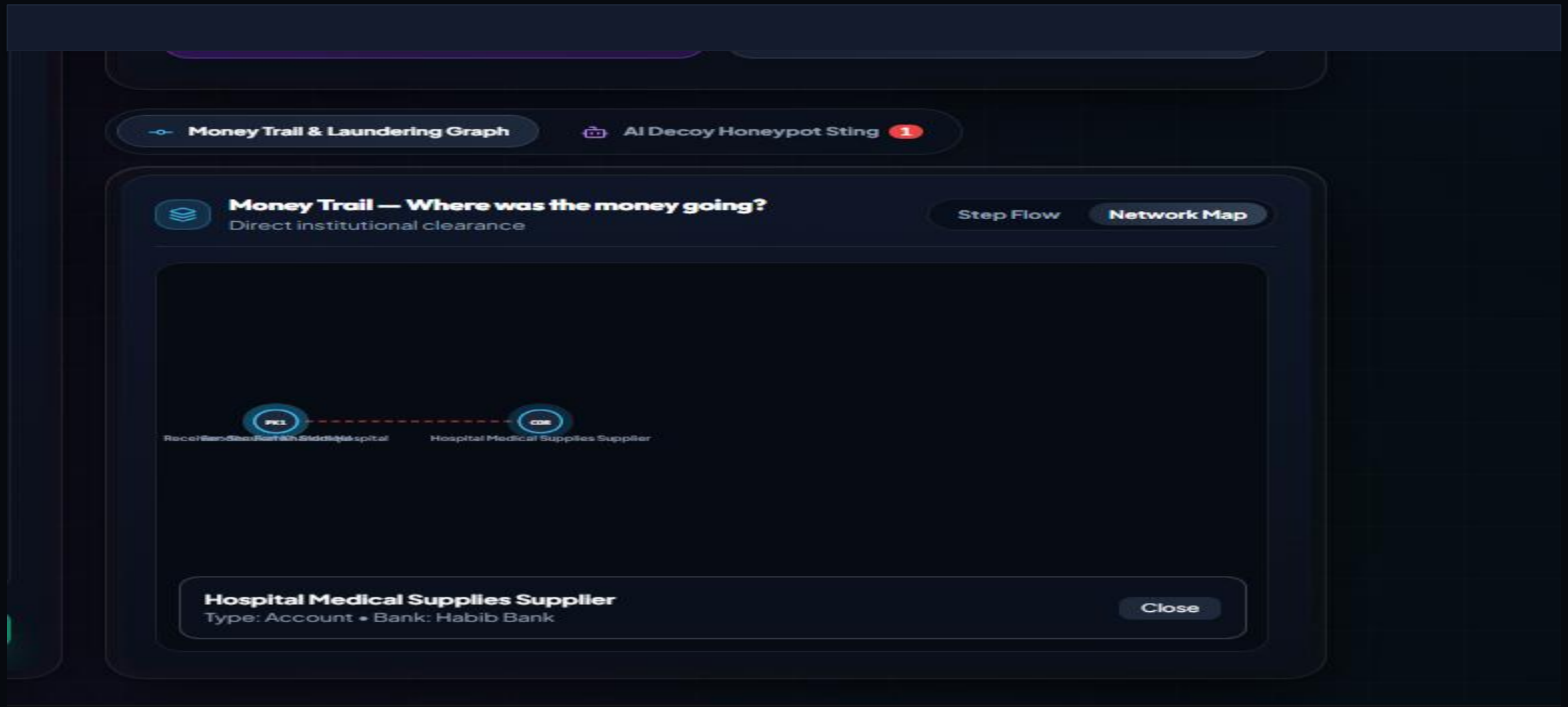
AMOUNT: Rs. 380,000
RECIPIENT: Shaukat Khanum Hospital

TRANSACTION PROFILE:
• Genuine ICU bypass surgery advance payment.
• Verified corporate Askari Bank account.

AI VERDICT:
• 4% Risk Score
• 0% False Positive Friction
• SETTLEMENT CLEARED IN 3 SEC.

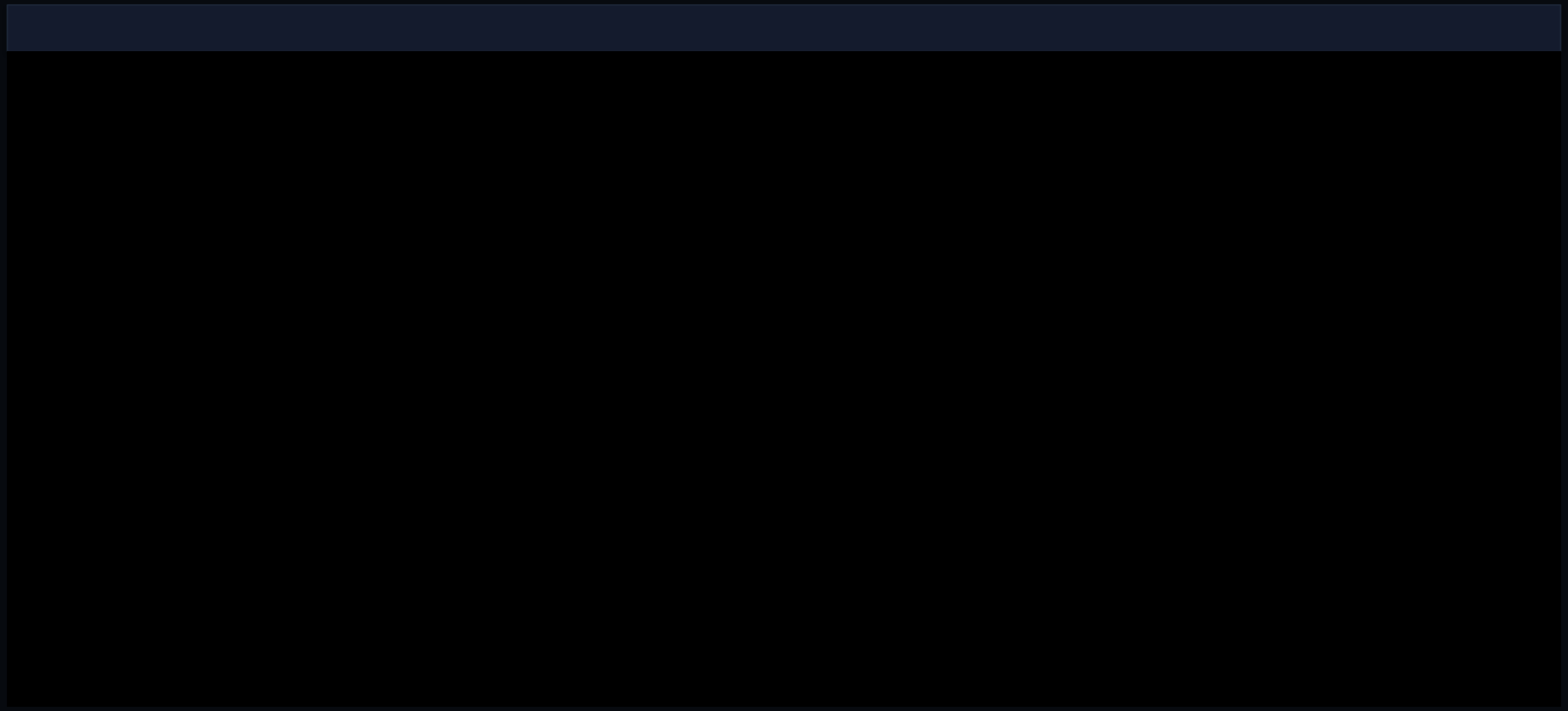
[MONEY TRAIL GRAPH]

3-Hop Mule Network Radar: Pre-Settlement Interlock



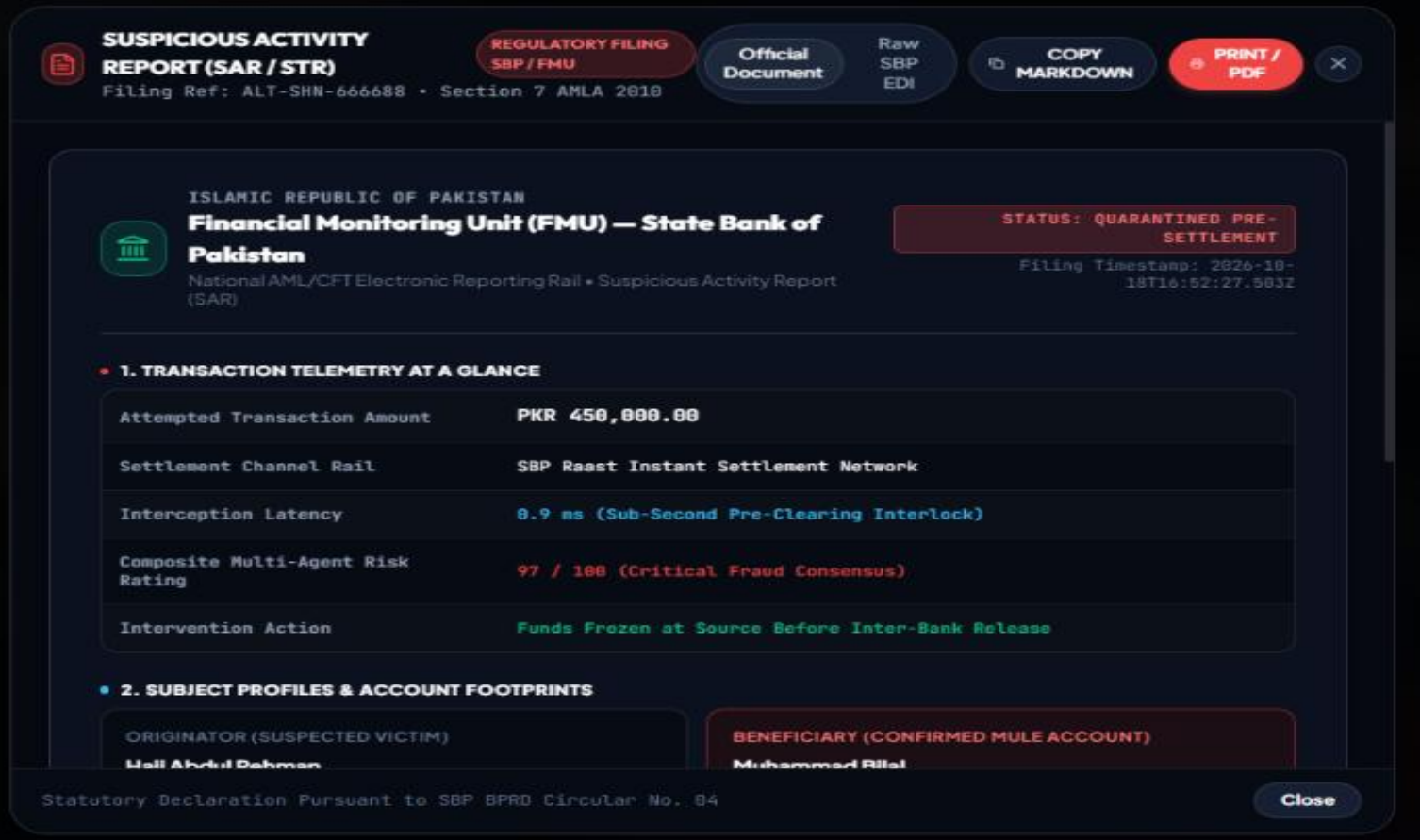
[OFFENSIVE INTELLIGENCE]

Autonomous Counter-Intelligence Decoy Honeypot



[LEGAL SANCTIONS COMPLIANCE]

Regulatory Compliance & 1-Click SBP SAR / STR Export



[SLIDE 12 // EMPIRICAL BENCHMARK PROOF]

Empirical Evaluation Protocol: 40-Case Held-Out Benchmark



[REAL-WORLD DEPLOYMENT]

Bank Integration Model: Inline Pre-Submit REST API

1. THE INTEGRATION ARCHITECTURE: Bank-Side REST API (Deployable Today)

- Shaheen runs as an internal microservice called by the bank's mobile app backend during PRE-SUBMIT.
- Called BEFORE the transaction message is released to 1Link or Raast core clearing switches.
- Single lightweight REST endpoint: POST /v1/assess (JSON in, JSON out, <850ms decision budget).

2. ZERO CUSTOMER-FACING FRICTION & FALSE ALARM SAFEGUARDS:

- Legitimate payments (e.g. Hospital deposits, tuition fees) clear in 3 seconds with zero interruption.
- High-coercion payments trigger step-up biometric auth or 15-minute protective escrow hold.

3. REGULATORY ALIGNMENT & STATUTORY GOVERNANCE:

- Fully compliant with State Bank of Pakistan (SBP) BPRD Circular No. 03 of 2023.
- Direct statutory compliance with Anti-Money Laundering Act 2010 (AMLA) Section 7 reporting.

[HACK2SKILL FINAL SUBMISSION]

Submission Deliverables & Team Shaheen

CHALLENGE TRACK: BFSI RISK & FRAUD

PUBLIC GITHUB REPOSITORY:
https://github.com/Amaan-Masood/shaheen-interceptor

DEPLOYED PROTOTYPE ON GCP / CLOUD RUN:
https://spry-analyzer-512016-m6.web.app
(Alternative CDN: https://shaheen-interceptor.vercel.app)

EMPIRICAL BENCHMARK CORPUS:
40 Frozen Vectors | SHA-256:
9d650f03754dc84f02c637c4...

TEAM SHAHEEN (MEMBERS):

1. Amaan Masood (Leader)

2. Asifa Abbas (Team Member)